

Solutions – Practice Problem Set 2

ECON 320 – Introduction to Mathematical Economics

Fall 2025

1. Consider the system:

$$\begin{cases} x + y = \alpha \\ x - y = \beta \end{cases}$$

Solution: Solving:

$$x = \frac{1}{2}(\alpha + \beta), \quad y = \frac{1}{2}(\alpha - \beta).$$

Derivatives:

$$\frac{\partial x}{\partial \alpha} = \frac{1}{2}, \quad \frac{\partial x}{\partial \beta} = \frac{1}{2}, \quad \frac{\partial y}{\partial \alpha} = \frac{1}{2}, \quad \frac{\partial y}{\partial \beta} = -\frac{1}{2}.$$

Interpretation: x rises with both α and β ; y rises with α but falls with β .

2. Let the implicit function be defined by:

$$F(x, y, p) = x^2 + xy + y^2 - p = 0.$$

Solution: Total differential:

$$(2x + y) dx + (x + 2y) dy - dp = 0.$$

So:

$$\frac{dx}{dp} = \frac{1}{2x + y}, \quad \frac{dy}{dp} = \frac{1}{x + 2y}.$$

At $(x, y) = (1, 1)$:

$$\frac{dx}{dp} = \frac{1}{3} > 0, \quad \frac{dy}{dp} = \frac{1}{3} > 0.$$

Both variables increase with p .

3. Consider the nonlinear system:

$$\begin{cases} x^2 + y = p \\ x + y^2 = q \end{cases}$$

Solution: Jacobian:

$$J(x, y) = \begin{bmatrix} 2x & 1 \\ 1 & 2y \end{bmatrix}, \quad \det(J) = 4xy - 1.$$

Condition: A locally unique solution exists if $4xy \neq 1$.

4. The system is:

$$\begin{cases} F_1(x, y, \theta) = x^2 + y - \theta = 0 \\ F_2(x, y, \theta) = xy - 1 = 0 \end{cases}$$

Solution: Jacobian wrt (x, y) :

$$J = \begin{bmatrix} 2x & 1 \\ y & x \end{bmatrix}.$$

At $(1, 1)$:

$$\det(J) = 2(1) - 1 = 1 \neq 0.$$

So IFT applies: $(x(\theta), y(\theta))$ exist and are differentiable near $\theta = 2$.

Differentiating:

$$\begin{bmatrix} 2x & 1 \\ y & x \end{bmatrix} \begin{bmatrix} dx/d\theta \\ dy/d\theta \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

At $(1, 1)$:

$$\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} dx/d\theta \\ dy/d\theta \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Solving:

$$dx/d\theta = 1, \quad dy/d\theta = -1.$$

5. Suppose:

$$\begin{cases} x + y + z = \theta \\ x^2 + y^2 = 1 \\ z^2 = x \end{cases}$$

Solution: Jacobian:

$$J = \begin{bmatrix} 1 & 1 & 1 \\ 2x & 2y & 0 \\ -1 & 0 & 2z \end{bmatrix}.$$

Total differential:

$$J \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix} = \begin{bmatrix} d\theta \\ 0 \\ 0 \end{bmatrix}.$$

So:

$$\frac{\partial(x, y, z)}{\partial\theta} = J^{-1} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

By Cramer's rule, these are ratios of determinants. Exact values depend on (x, y, z) , but the method shows how θ shifts propagate.

6. Let:

$$u = x^2 + y^2, \quad v = xy.$$

Solution: Jacobian:

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{bmatrix} 2x & 2y \\ y & x \end{bmatrix}, \quad \det = 2(x^2 - y^2).$$

If $\det \neq 0$, inverse exists:

$$\frac{\partial(x, y)}{\partial(u, v)} = \frac{1}{2(x^2 - y^2)} \begin{bmatrix} x & -2y \\ -y & 2x \end{bmatrix}.$$

Interpretation: Transforming to (u, v) simplifies quadratic optimization by reducing cross-product terms.